Technical Report: CodeIgniter 4 CRUD System

Product Management & POS System

Project: CI4_BSIT3A | **Framework:** CodeIgniter 4 | **Database:** MySQL | **Date:** December 10, 2025

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1. System Overview

Web-based application for small to medium businesses providing inventory management, POS operations, user authentication, and sales reporting with QR code integration and SMS notifications.

Key Features: Product CRUD, QR code generation, POS system, user authentication (bcrypt), transaction management, soft delete, search/pagination, mobile-responsive design.

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2. System Architecture

MVC Structure

```
ci4_BSIT3A/
├── Controllers/    (Auth, Product, POS, User)
├── Models/         (Product, User, Transaction, TransactionItem)
├── Views/          (Product, POS, User Portal)
├── Libraries/      (QrCodeService, SmsService)
└── Config/         (Routes, Database, Filters)
```

Controllers

AuthController - Authentication

```php
namespace App\Controllers;
use App\Models\UserModel;

class AuthController extends Controller
{
    protected $userModel;

    public function login() { }              // Display login form
    public function processLogin() { }       // Process login credentials
    public function register() { }           // Display registration form
    public function processRegister() { }    // Process registration
    public function logout() { }             // Logout user
    public function checkAuth() { }          // Check auth status (AJAX)
}
```

ProductController - Product Management

```php
namespace App\Controllers;
use App\Models\ProductModel;

class ProductController extends BaseController
{
    protected $productModel;

    public function index() { }              // List products with pagination
    public function showCreateForm() { }     // Display create form
    public function create() { }             // Create new product
    public function edit($id) { }            // Display edit form
    public function update($id) { }          // Update product
    public function view($id) { }            // View product details
    public function delete($id) { }          // Soft delete product
    public function restore($id) { }         // Restore deleted product
    public function permanentDelete($id) { } // Permanently delete
    public function trash() { }              // View deleted products
    public function bulkDelete() { }         // Bulk delete operations
    public function bulkRestore() { }        // Bulk restore operations
    public function statistics() { }         // Get statistics (API)
}
```

PosController - Point of Sale

```php
namespace App\Controllers;

class PosController extends BaseController
{
    public function dashboard() { }          // POS dashboard with statistics
    public function scanner() { }            // QR code scanner interface
    public function processScan() { }        // Process scanned QR code
    public function updateCart() { }         // Update cart items
    public function getCart() { }            // Get cart contents (AJAX)
    public function cartCount() { }          // Get cart count (AJAX)
    public function checkout() { }           // Display checkout page
    public function processCheckout() { }    // Process transaction
    public function receipt($id) { }         // Display receipt
    public function resendSms($id) { }       // Resend SMS receipt
    public function searchProducts() { }     // Search products (AJAX)
    public function addProduct() { }         // Add product to cart
}
```

UserController - User Portal

```php
namespace App\Controllers;
```

```
class UserController extends Controller
{
    public function dashboard() { }          // User dashboard
    public function profile() { }            // User profile page
    public function updateProfile() { }      // Update profile
    public function transactions() { }       // Transaction history
    public function viewTransaction($id) { } // View transaction details
    public function products() { }           // Browse products
    public function viewProduct($id) { }     // View product details
}
```

Models

ProductModel

```
namespace App\Models;
use CodeIgniter\Model;

class ProductModel extends Model
{
    protected $table = 'products';
    protected $primaryKey = 'id';
    protected $allowedFields = ['product_name', 'description', 'price', 'qr_code',
'stock_quantity', 'status', 'product_image'];
    protected $useTimestamps = true;

    protected $validationRules = [
        'product_name' => 'required|min_length[3]|max_length[255]',
        'price' => 'required|decimal|greater_than[0]',
        'stock_quantity' => 'permit_empty|integer|greater_than_equal_to[0]'
    ];

    public function createProduct($data) { }
    public function getProduct($id) { }
    public function getProductsPaginated($perPage = 10) { }
    public function updateProduct($id, $data) { }
    public function deleteProduct($id) { }
    public function restoreProduct($id) { }
    public function permanentlyDeleteProduct($id) { }
    public function searchProducts($keyword, $perPage) { }
    public function generateQrCode($productId) { }
}
```

UserModel

```
namespace App\Models;
use CodeIgniter\Model;

class UserModel extends Model
```

```
{
    protected $table = 'users';
    protected $primaryKey = 'id';
    protected $allowedFields = ['username', 'password'];
    protected $useTimestamps = true;

    protected $validationRules = [
        'username' =>
'required|min_length[3]|max_length[50]|is_unique[users.username]',
        'password' => 'required|min_length[6]|max_length[255]'
    ];

    public function createUser(array $data) { }
    public function authenticateUser(string $username, string $password) { }
    public function getUserByUsername(string $username) { }
}
```

TransactionModel

```
namespace App\Models;
use CodeIgniter\Model;

class TransactionModel extends Model
{
    protected $table = 'transactions';
    protected $primaryKey = 'id';
    protected $allowedFields = ['transaction_number', 'total_amount',
'payment_method', 'customer_phone', 'sms_sent', 'receipt_data'];
    protected $useTimestamps = true;

    public function createTransaction($data) { }
    public function getTransactionWithItems($transactionId) { }
    public function getDailySales($date = null) { }
    public function getRecentTransactions($limit) { }
}
```

TransactionItemModel

```
namespace App\Models;
use CodeIgniter\Model;

class TransactionItemModel extends Model
{
    protected $table = 'transaction_items';
    protected $primaryKey = 'id';
    protected $allowedFields = ['transaction_id', 'product_id', 'product_name',
'price', 'quantity', 'subtotal'];
    protected $useTimestamps = true;
```

```
    public function addItem($data) { }
    public function getTransactionItems($transactionId) { }
    public function getTotalQuantitySold($productId) { }
}
```

Libraries

QrCodeService

```
namespace App\Libraries;
use BaconQrCode\Renderer\ImageRenderer;
use BaconQrCode\Renderer\Image\SvgImageBackEnd;
use BaconQrCode\Renderer\RendererStyle\RendererStyle;
use BaconQrCode\Writer;

class QrCodeService
{
    protected $writer;
    protected $renderer;

    public function __construct()
    {
        $this->renderer = new ImageRenderer(new RendererStyle(400), new
SvgImageBackEnd());
        $this->writer = new Writer($this->renderer);
    }

    public function generateProductQrCode($productId, $productName, $price) { }
    public function generateProductQrCodeDataUri($productId, $productName, $price)
{ }
    public function saveProductQrCodeToFile($productId, $productName, $price,
$filePath) { }
}
```

SmsService

```
namespace App\Libraries;
use CodeIgniter\HTTP\CURLRequest;
use Config\Services;

class SmsService
{
    protected $apiToken = '1ef3b27ea753780a90cbdf07d027fb7b52791004';
    protected $apiUrl = 'https://sms.iprogtech.com/api/v1/sms_messages';
    protected $client;

    public function __construct()
    {
        $this->client = Services::curlrequest();
```

```
    }

    public function sendSms($phoneNumber, $message) { }
    public function sendReceiptSms($phoneNumber, $transactionData) { }
    public function validatePhoneNumber($phoneNumber) { }
}
```

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3. Routing Configuration

File: app/Config/Routes.php

```
<?php
use CodeIgniter\Router\RouteCollection;

// DEFAULT ROUTE
$routes->get('/', 'ProductController::index');

// AUTHENTICATION ROUTES
$routes->group('auth', function ($routes) {
    $routes->get('login', 'AuthController::login');
    $routes->post('processLogin', 'AuthController::processLogin');
    $routes->get('register', 'AuthController::register');
    $routes->post('processRegister', 'AuthController::processRegister');
    $routes->get('logout', 'AuthController::logout');
    $routes->get('check', 'AuthController::checkAuth');
});

// Legacy login routes
$routes->get('login', 'AuthController::login');
$routes->post('login', 'AuthController::processLogin');
$routes->get('logout', 'AuthController::logout');

// USER PORTAL ROUTES
$routes->group('user', function ($routes) {
    $routes->get('dashboard', 'UserController::dashboard');
    $routes->get('profile', 'UserController::profile');
    $routes->post('update-profile', 'UserController::updateProfile');
    $routes->get('transactions', 'UserController::transactions');
    $routes->get('transaction/(:num)', 'UserController::viewTransaction/$1');
    $routes->get('products', 'UserController::products');
    $routes->get('product/(:num)', 'UserController::viewProduct/$1');
});

// POS SYSTEM ROUTES
$routes->group('pos', function ($routes) {
    $routes->get('dashboard', 'PosController::dashboard');
    $routes->get('scanner', 'PosController::scanner');
    $routes->post('process-scan', 'PosController::processScan');
    $routes->post('update-cart', 'PosController::updateCart');
    $routes->get('cart', 'PosController::getCart');
```

```
    $routes->get('cart-count', 'PosController::cartCount');
    $routes->get('checkout', 'PosController::checkout');
    $routes->post('process-checkout', 'PosController::processCheckout');
    $routes->get('receipt/(:num)', 'PosController::receipt/$1');
    $routes->post('resend-sms/(:num)', 'PosController::resendSms/$1');
    $routes->post('search-products', 'PosController::searchProducts');
    $routes->post('add-product', 'PosController::addProduct');
    $routes->post('check-customer', 'PosController::checkCustomer');
});

// PRODUCT CRUD ROUTES
$routes->get('products', 'ProductController::index');
$routes->get('products/create', 'ProductController::showCreateForm');
$routes->post('products/create', 'ProductController::create');
$routes->get('products/view/(:num)', 'ProductController::view/$1');
$routes->get('products/edit/(:num)', 'ProductController::edit/$1');
$routes->post('products/update/(:num)', 'ProductController::update/$1');
$routes->get('products/delete/(:num)', 'ProductController::delete/$1');

// IMAGE SERVING ROUTE
$routes->get('writable/uploads/(:any)', 'ProductController::serveImage/$1');

// TRASH MANAGEMENT ROUTES
$routes->get('trash', 'ProductController::trash');
$routes->get('products/restore/(:num)', 'ProductController::restore/$1');
$routes->get('products/permanent-delete/(:num)',
'ProductController::permanentDelete/$1');

// BULK OPERATIONS ROUTES
$routes->post('products/bulk-delete', 'ProductController::bulkDelete');
$routes->post('products/bulk-restore', 'ProductController::bulkRestore');
$routes->post('products/bulk-permanent-delete',
'ProductController::bulkPermanentDelete');

// API ROUTES
$routes->get('api/statistics', 'ProductController::statistics');

// LEGACY ROUTES
$routes->get('home', 'Home::index');
$routes->get('lecture', 'Home::lectureNotes');
```

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4. Database Design

Tables

- **products**: id, product_name, description, price, qr_code, stock_quantity, status, product_image, created_at, updated_at
- **users**: id, username, password, created_at, updated_at
- **transactions**: id, transaction_number, total_amount, payment_method, customer_phone, sms_sent, receipt_data, created_at, updated_at

- **transaction_items**: id, transaction_id, product_id, product_name, price, quantity, subtotal, created_at, updated_at

Relationships

- USERS (1) → (M) PRODUCTS
- PRODUCTS (1) → (M) TRANSACTION_ITEMS
- TRANSACTIONS (1) → (M) TRANSACTION_ITEMS

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5. Authentication & Security

Admin Account: username: `admin` | password: `admin123`

Security Features:

- Bcrypt password hashing (cost factor 12)
- CSRF protection with token validation
- Input validation and sanitization
- SQL injection prevention (Query Builder)
- XSS protection (output escaping)
- Session security (HTTPOnly, 7200s timeout)
- File upload restrictions (jpg, png, gif, max 2MB)

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6. API Integration

QR Code API: BaconQrCode v6.0 - Generates SVG QR codes with product data (ID, name, price, timestamp)

SMS Service: iProg Tech API - Sends transaction receipts to customers

Endpoints:

- `GET /products?search={keyword}` - Search products
- `POST /pos/process-scan` - Process QR code
- `POST /pos/process-checkout` - Create transaction
- `GET /api/statistics` - Dashboard statistics

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7. System Features

Product Management: Create, Read, Update, Delete, Search, Pagination, Soft Delete, Restore, Bulk Operations

POS System: QR Scanner, Shopping Cart, Checkout, Multiple Payment Methods, SMS Receipts, Receipt Printing, Statistics

User Management: Registration, Login, Profile Management, Password Change

Reporting: Sales Analytics, Inventory Tracking, Export (PDF/CSV)

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8. Testing & Validation

☑ User authentication (login, registration, logout) ☑ Product CRUD operations ☑ QR code generation and scanning ☑ Transaction processing ☑ Search and pagination ☑ Password hashing verification ☑ CSRF protection ☑ SQL injection prevention ☑ XSS prevention ☑ Page load time < 2 seconds ☑ Mobile responsiveness

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9. System Setup

Prerequisites: PHP 8.1+, Apache/Nginx, MySQL 5.7+, Composer

Installation:

```
git clone https://github.com/Jan-Russel-Penafiel/CI4_CRUD.git
cd ci4_BSIT3A
composer install
CREATE DATABASE ci4_bsit3a;
cp env .env
# Edit .env with database credentials
chmod -R 755 writable/
php spark serve
# Access at http://localhost:8080
```

Dependencies: PHP 8.1+, CodeIgniter 4 ^4.0, BaconQrCode ^6.0, PHPUnit ^10.5.16

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10. Screenshots

Authentication

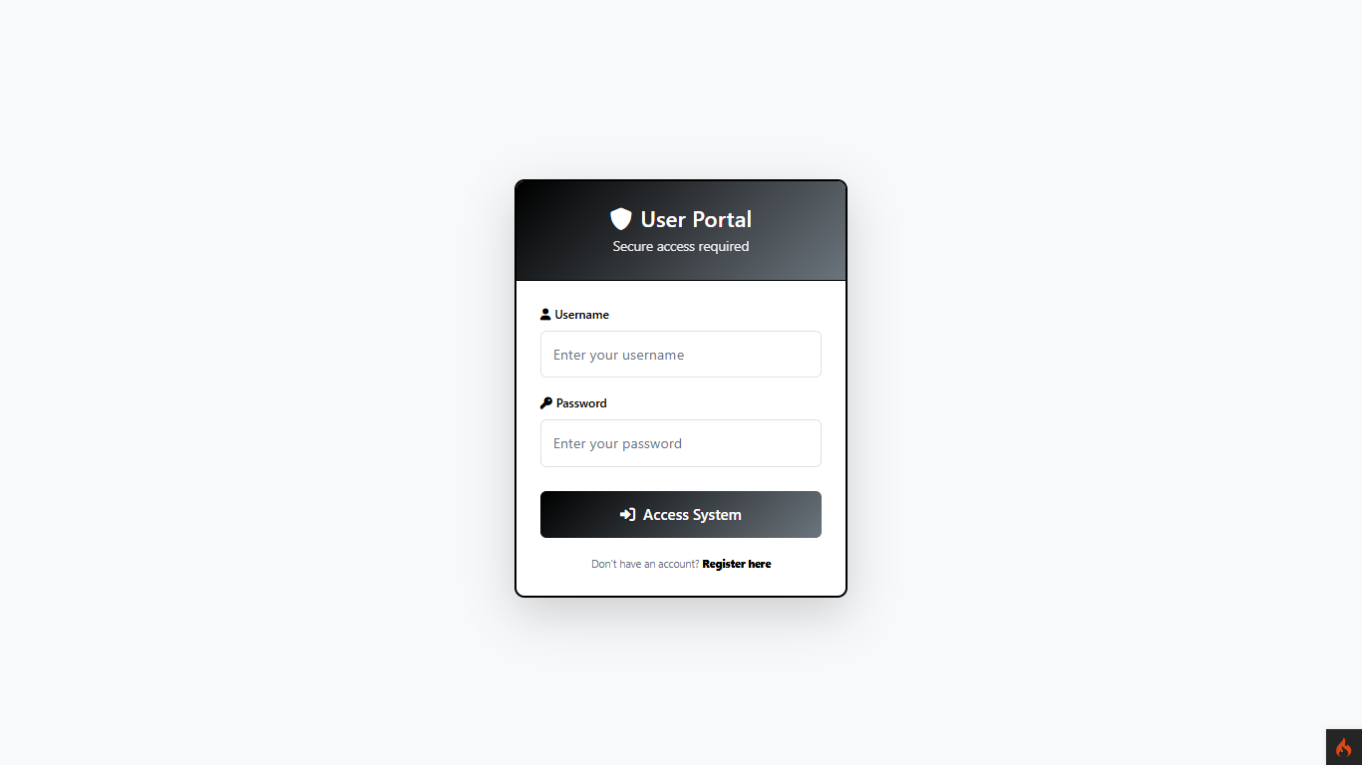


Figure 1: Login interface with CSRF protection

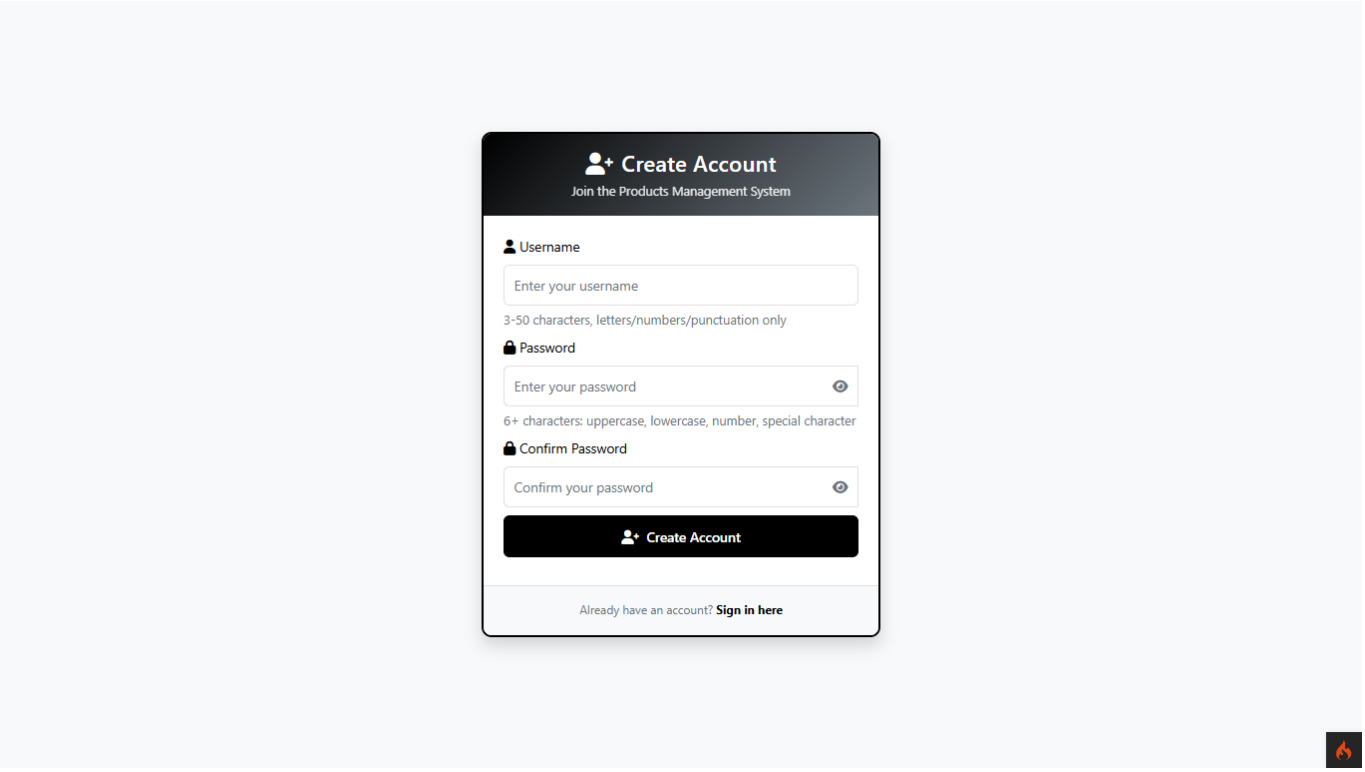


Figure 2: Registration form with password strength validation

Product Management

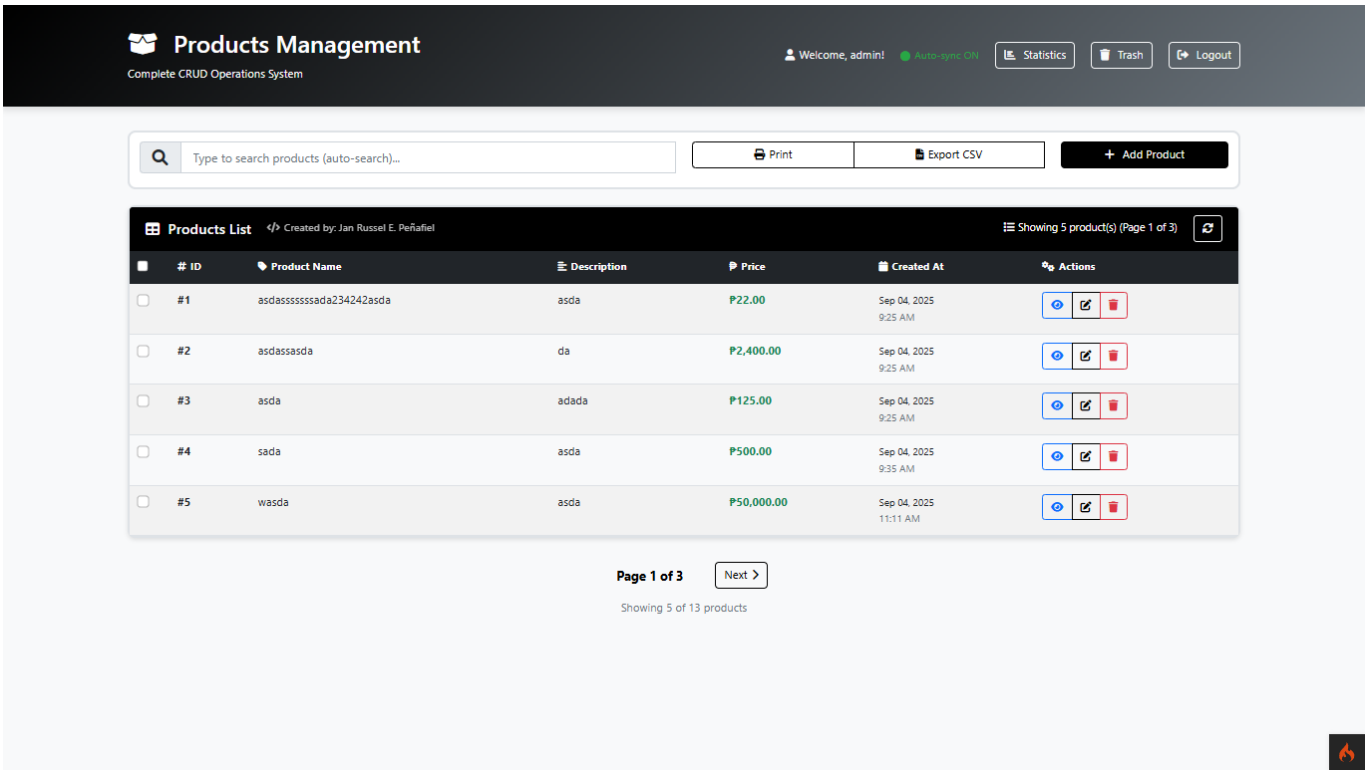


Figure 3: Product listing with search and pagination

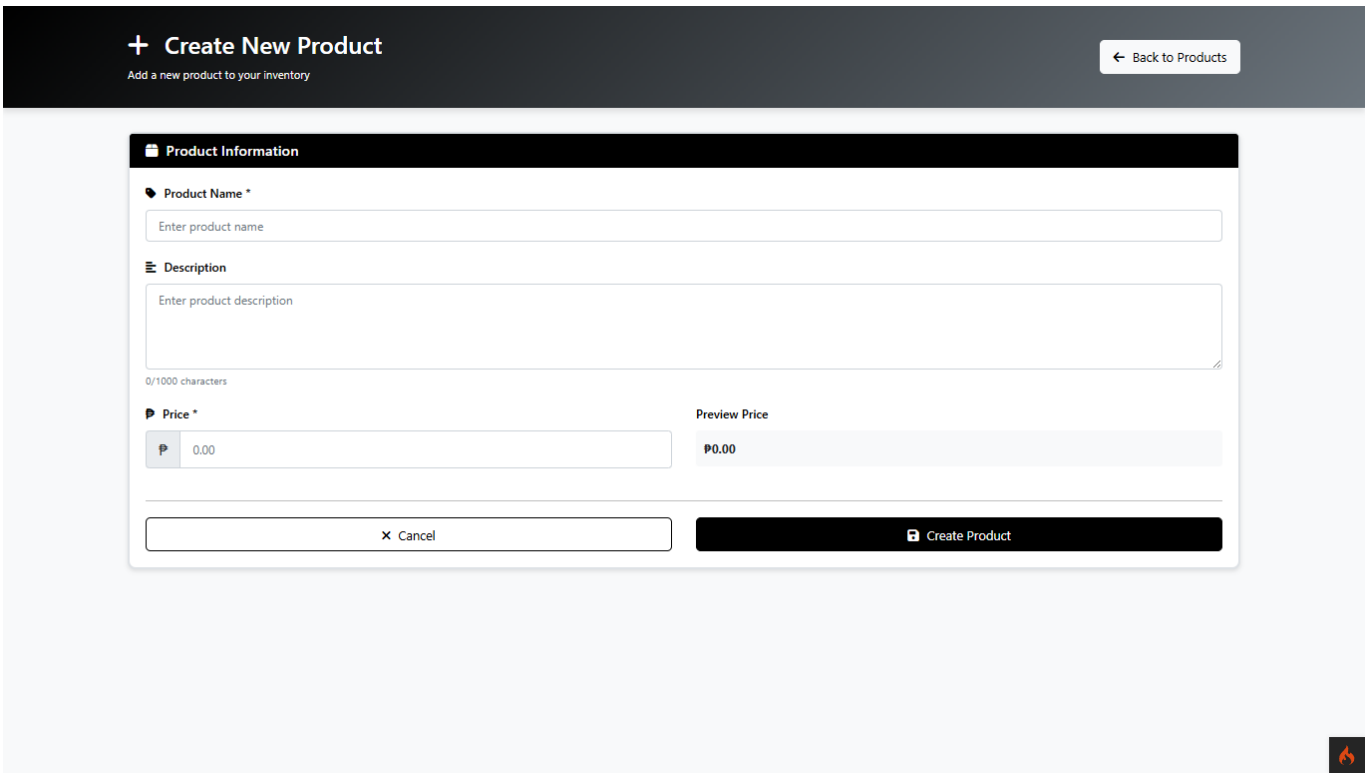


Figure 4: Product creation with validation

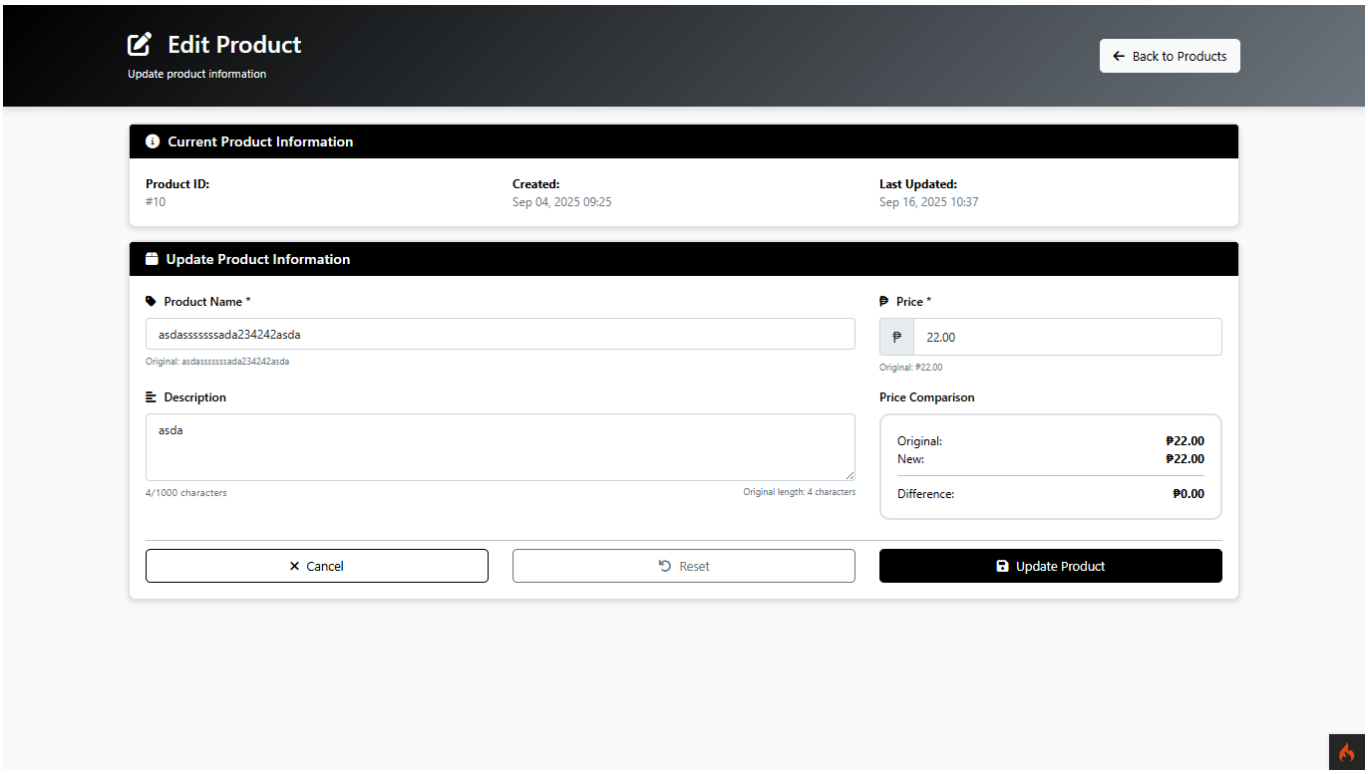


Figure 5: Product editing interface

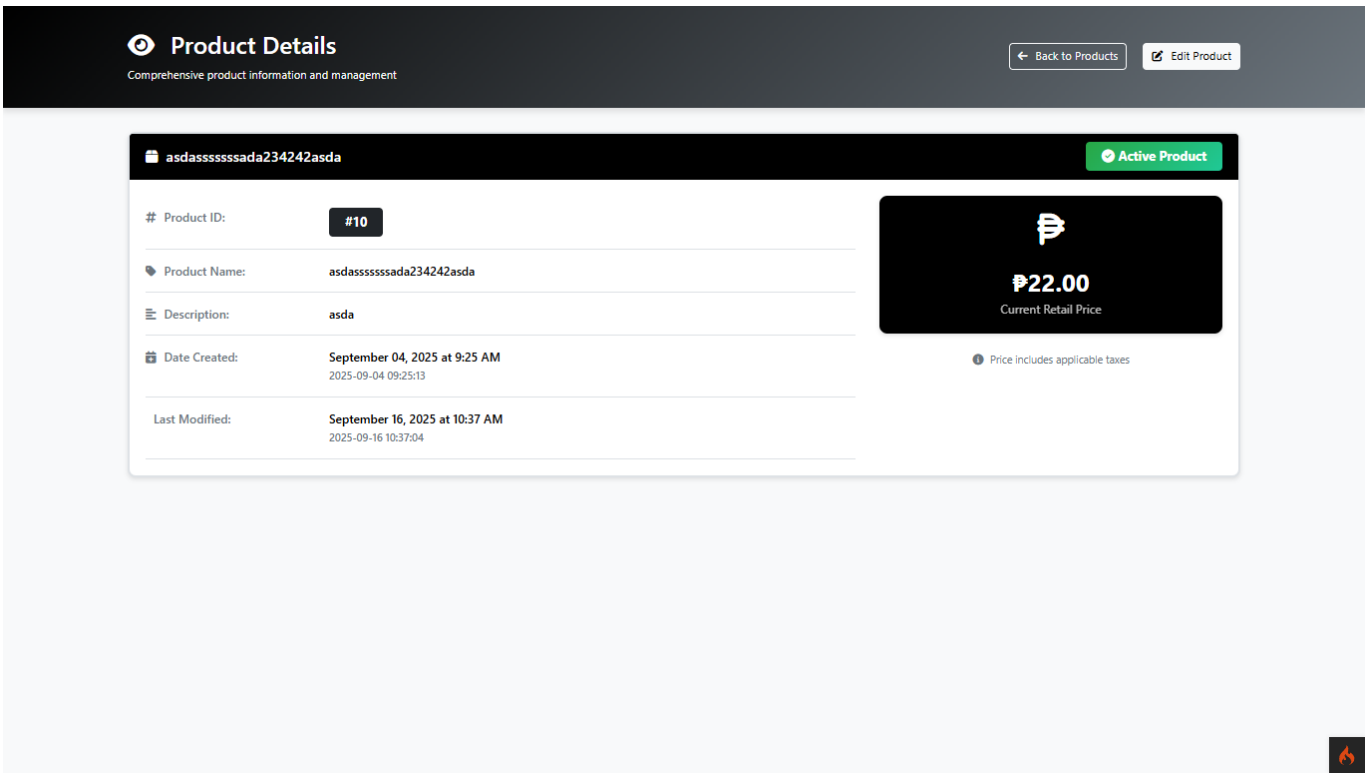


Figure 6: Product details with QR code

Advanced Features

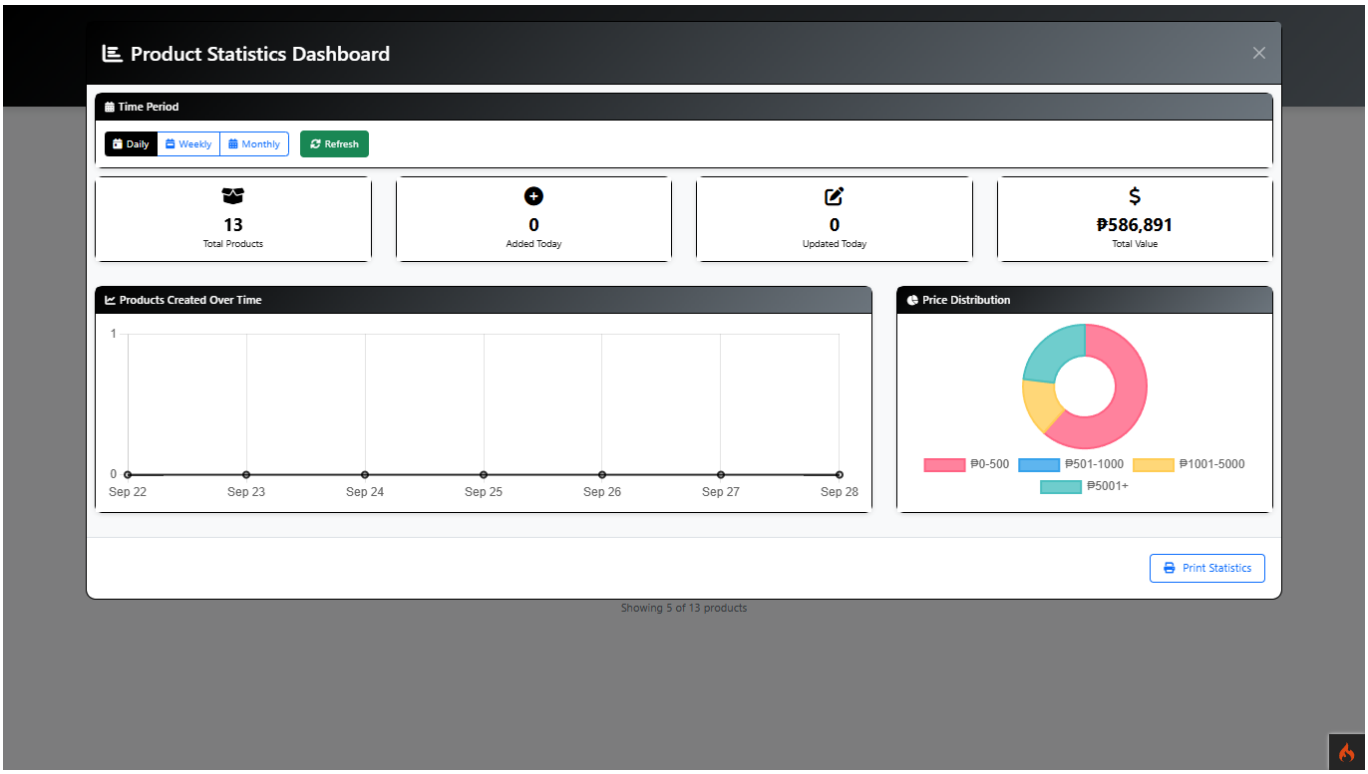


Figure 7: Analytics dashboard

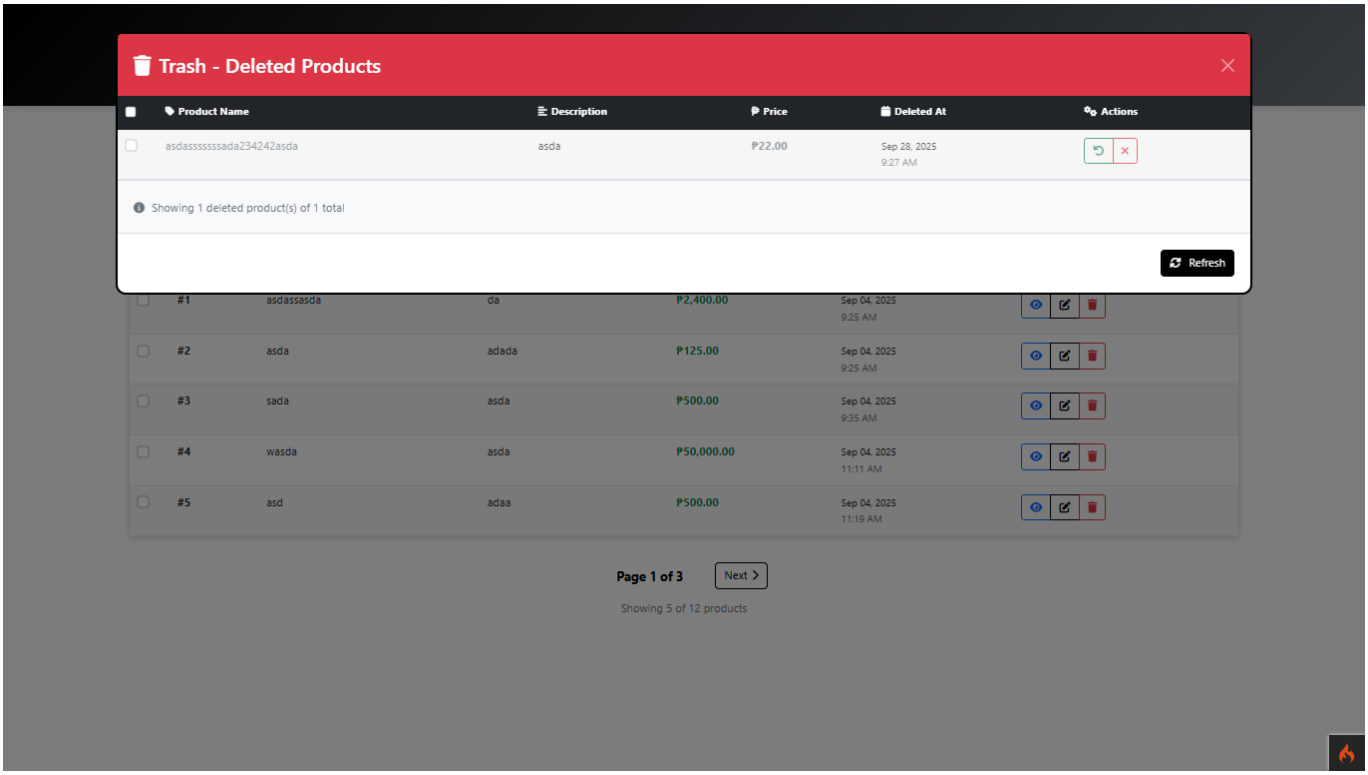


Figure 8: Soft delete management

Point-of-Sale System

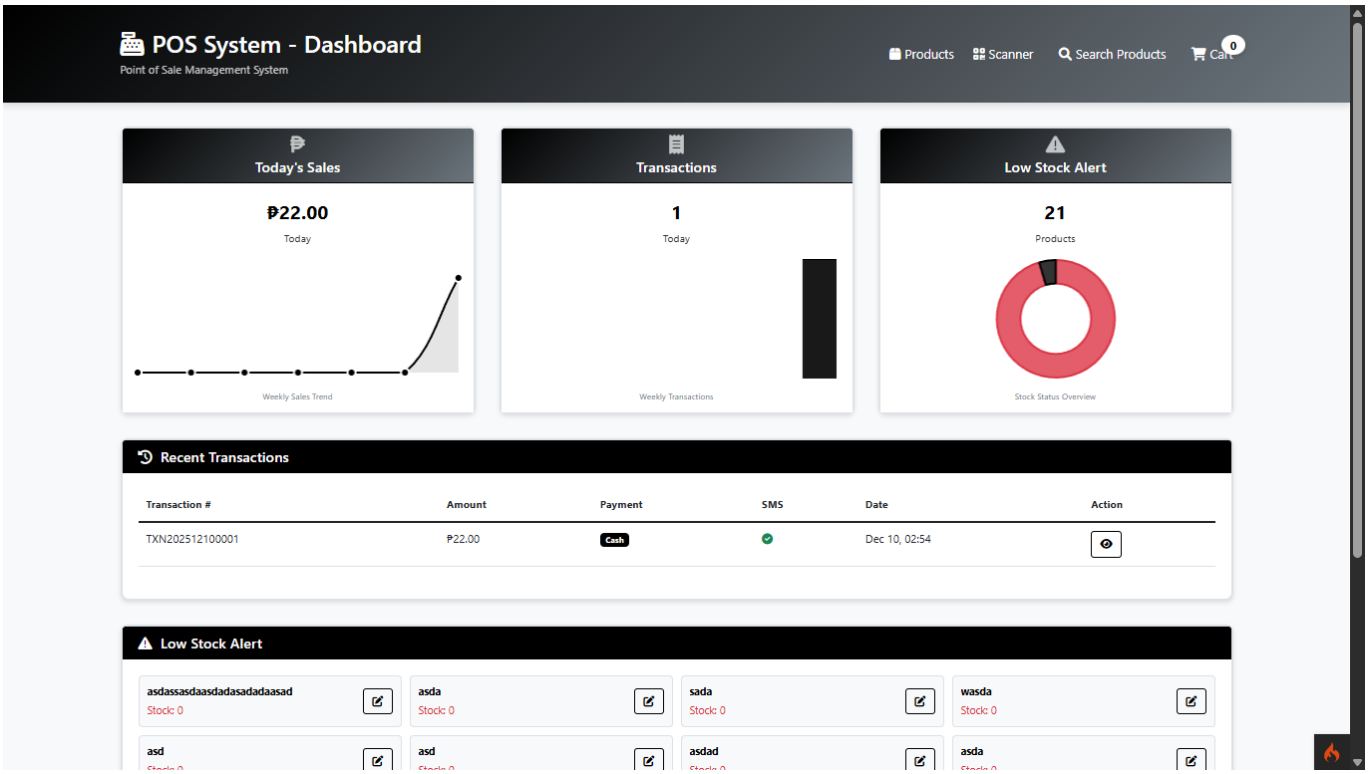


Figure 9: POS dashboard with statistics

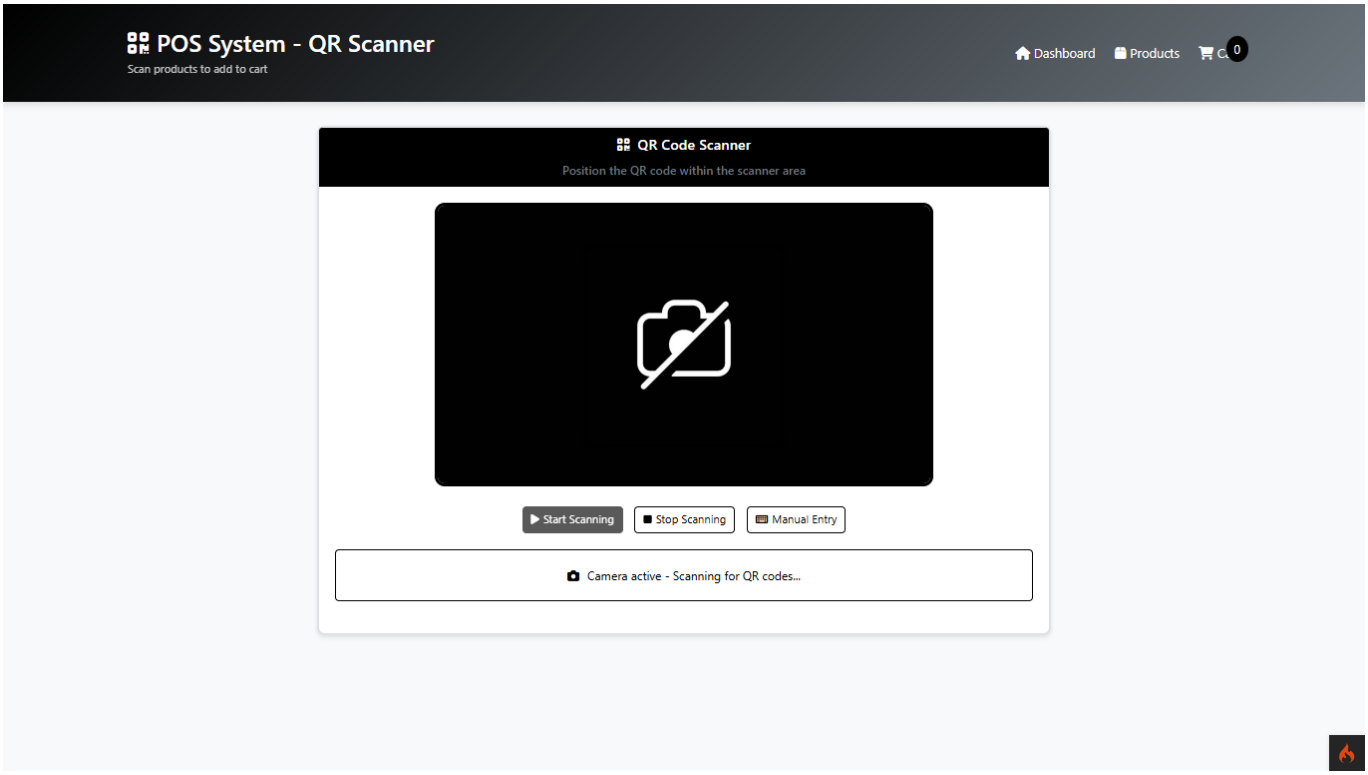


Figure 10: QR code scanner interface

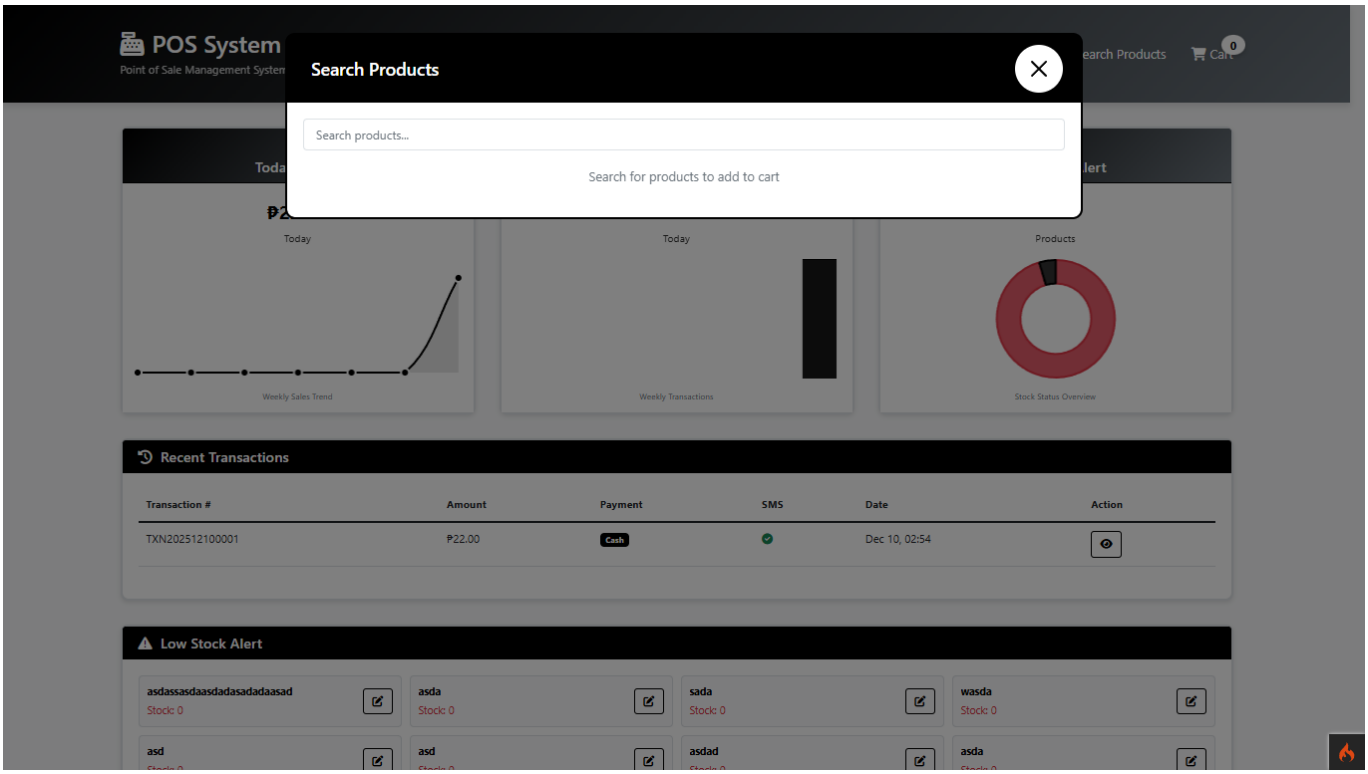


Figure 11: Product search in POS

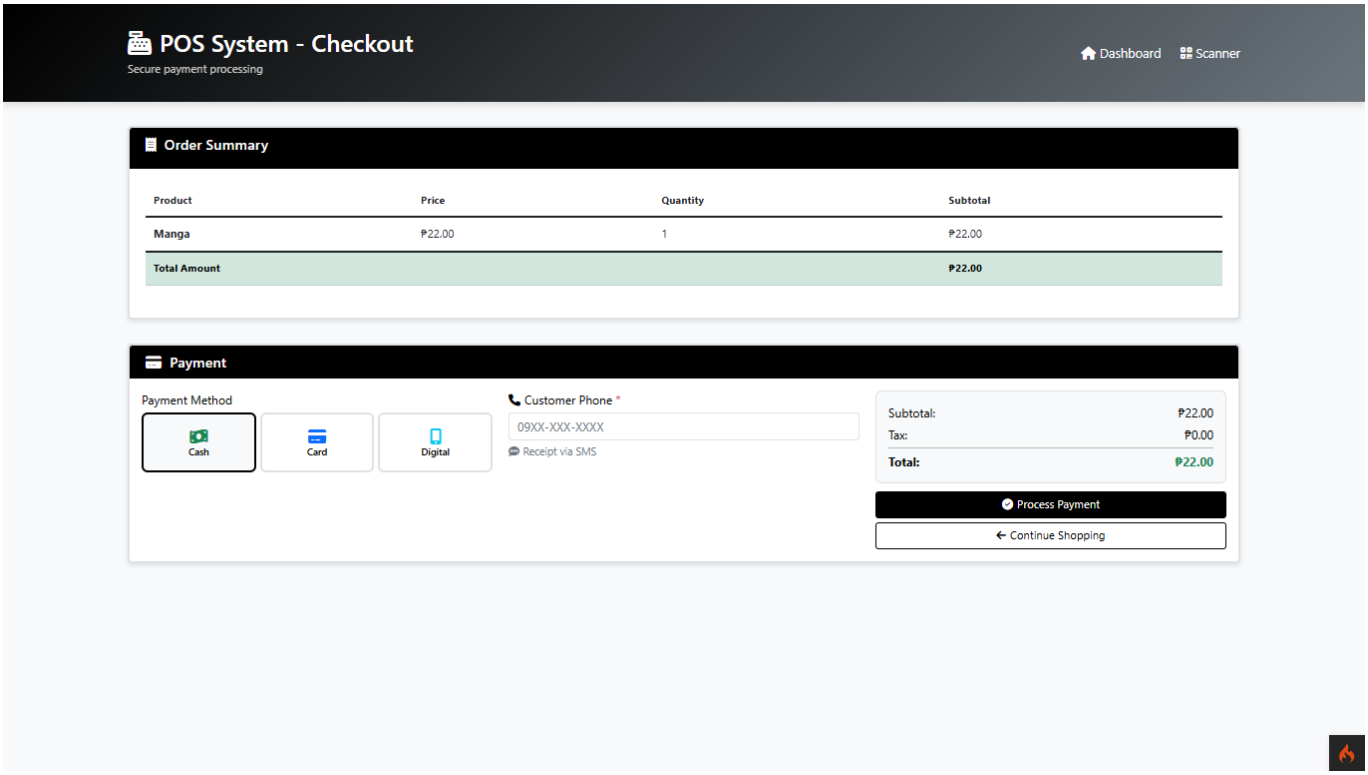


Figure 12: Checkout interface

User Portal

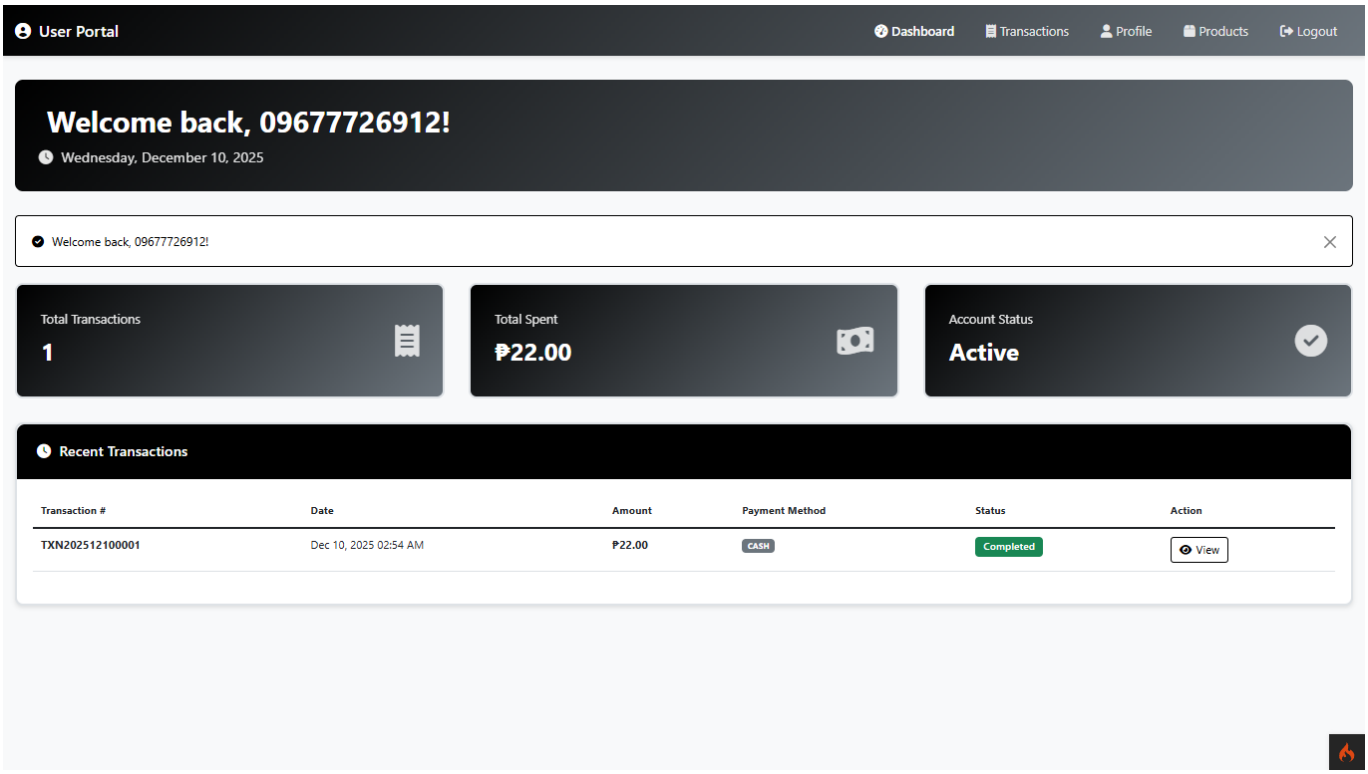


Figure 13: User dashboard

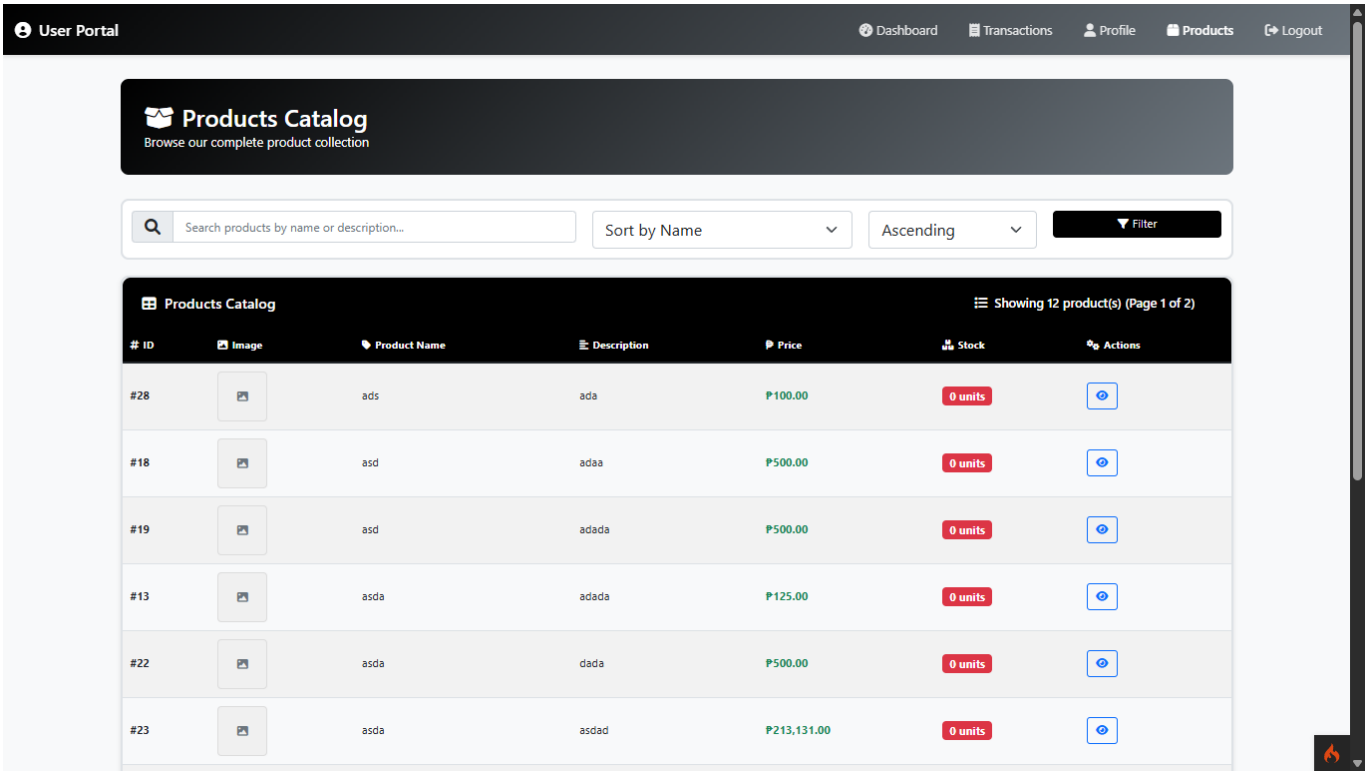


Figure 14: Product browsing

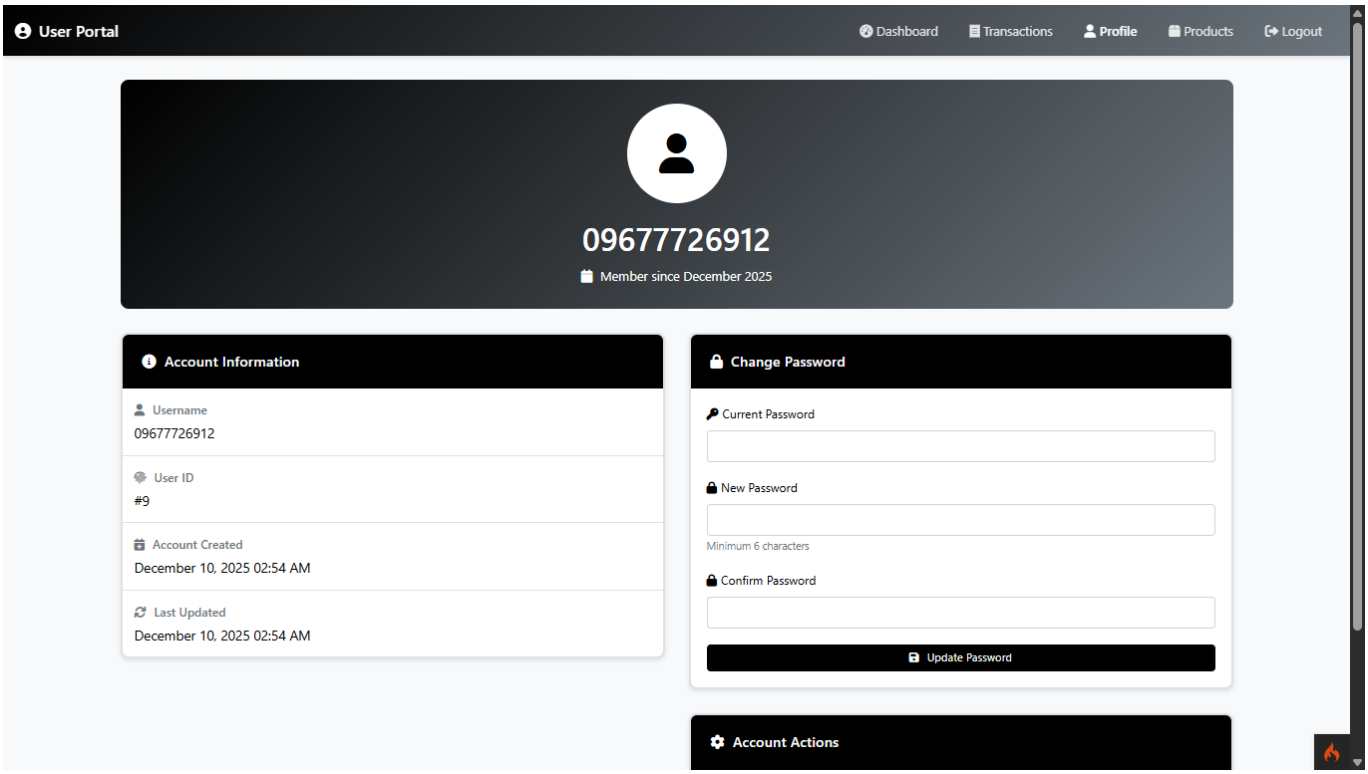


Figure 15: User profile management

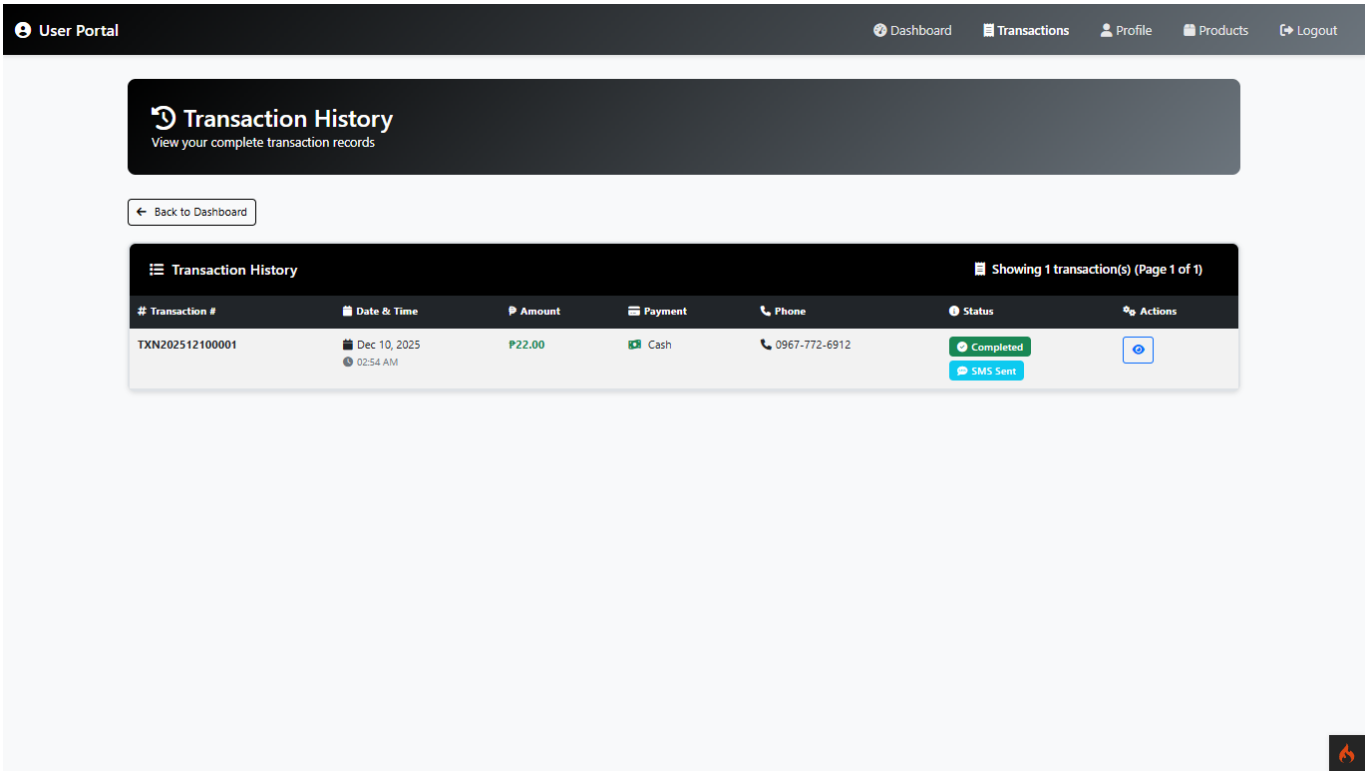


Figure 16: Transaction history

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11. Conclusion

Strengths: Modern CodeIgniter 4, comprehensive security, scalable MVC architecture, responsive design, complete CRUD, production-ready

Limitations: Single database support, no native mobile app, limited real-time features, single language

Future Improvements: RESTful API, mobile apps, advanced analytics, multi-tenant support, AI integration, cloud migration

Summary: Production-ready system with 4 controllers, 4 models, 30+ routes, 4 database tables, complete CRUD operations, comprehensive security, and responsive UI with 16 screenshots.

Version: 2.0 | **Status:** Production-Ready | **Date:** December 10, 2025